Research Statement: Extending complex non-Abelian Hodge theory to moduli stacks

1 Background. The discontinuity of the non-Abelian Hodge (NAH) correspondence on the non-polystable locus of the Dolbeault moduli space [12, pp. 38–39] raised interest in finding an NAH correspondence between the respective moduli stacks [9, p. 12]. The Borel-Moore homologies of topological versions of these stacks are known to be isomorphic in the case of curves [4, 6]. In [1], we constructed diffeologically presentable stacks (in the sense of [8]) $\mathcal{M}_{\mathrm{Dol}}(X)$ and $\mathcal{M}_{\mathrm{dR}}(X)$, over the site of smooth manifolds, parametrizing smooth families of Higgs and flat bundles respectively. By relativising the extension completion of dg-categories from [11], we produced an equivalence of substacks $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol}}(X) \subset \mathcal{M}_{\mathrm{Dol}}(X)$ and $\mathcal{M}^{\mathcal{H}}_{\mathrm{dR}}(X) \subset \mathcal{M}_{\mathrm{dR}}(X)$. This equivalence is an extension of the classical equivalence of categories to moduli stacks that also accounts for parts of the non-polystable locus where the bijection of coarse moduli spaces becomes discontinuous. In [2], we showed that $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol}}(X)$ contains the stable locus but not the entire polystable locus. Nevertheless, the same results hold for larger moduli stacks $\mathcal{M}^{\mathcal{H}}_{\mathrm{Dol},d}(X) \subset \mathcal{M}_{\mathrm{Dol},d}(X)$ and $\mathcal{M}^{\mathcal{H}}_{\mathrm{dR},d}(X) \subset \mathcal{M}_{\mathrm{dR},d}(X)$ parametrizing analogous C^d families. Allowing harmonic metrics with discontinuities in the parameter, or weak harmonic families, gave an equivalence of still larger substacks $\mathcal{M}^{\mathrm{wk}\mathcal{H}}_{\mathrm{Dol},0}(X) \subset \mathcal{M}_{\mathrm{Dol},0}(X)$ and $\mathcal{M}^{\mathrm{wk}\mathcal{H}}_{\mathrm{dR},0}(X) \subset \mathcal{M}_{\mathrm{dR},0}(X)$. The pattern here is: weakening of regularity hypotheses gives extensions of the stack-level correspondence. However, it is not clear how far this goes, and I hope to investigate this phenomenon in my postdoctoral research. Please see this URL for a more detailed discussion.

2 Sporadic Harmonic Mediator. Consider this example due to my doctoral supervisor:

Example 2.1. For an elliptic curve X, consider the complex vector bundle $E = (\pi_X^* \mathcal{O}_X)^{\oplus 3} \cong \pi_X^*(\mathcal{O}_X^{\oplus 3})$ over $\mathbb{C} \times X$ obtained by pullback along the projection $\pi_X : \mathbb{C} \times X \longrightarrow X$. Let $\alpha \in H^0(X, \Omega^1_X)$ be a non-zero 1–form and:

$$\theta(z) = \begin{bmatrix} 0 & z & 0 \\ 0 & 0 & z \\ z^5 & 0 & 0 \end{bmatrix} \alpha$$

Then, (E, θ) is a holomorphic family of polystable Higgs bundles on X parametrized by $\mathbb{C}$.

We can show that for every choice of harmonic metric, the operator family $\{D_z\}_{z \in \mathbb{C}}$ obtained by applying the NAH transform to each slice $(E_z, \theta(z)), z \in \mathbb{C}$ is discontinuous at 0. Thus, even $\mathcal{M}^{\mathrm{wk}\mathcal{H}}_{\mathrm{Dol},0}(X)$ does not contain the entire polystable locus. This also implies that any mapping from Higgs bundle families to flat bundle families that applies the NAH transform slicewise cannot generate a continuous flat bundle family from Example 2.1, and hence cannot cover the entire polystable locus. Since the NAH correspondence is continuous on the coarse polystable locus, this is surprising but abstracting the properties of the problematic Higgs bundle family above suggests a way forward:

Definition 2.2 (Sporadic Families)**.** A tuple $(U, P, \{U_i\}_{i \in I}, E, \theta, \{D_i\}_{i \in I})$ where U is a C^0 manifold, $P \subset U$ is closed, $\{U_i\}_{i \in I}$ is an open cover of $U \setminus P$, and:

- $(E \longrightarrow U \times X, \theta)$ is a continuous family of Higgs bundles on X parametrized by U.
- $(E|(U_i \times X), D_i)$ is a continuous family of flat bundles on X parametrized by U_i.
- For each $i \in I, u \in U_i$, $(E_u, \theta_u, D_{i,u})$ is a harmonic bundle.
- For each $i, j \in I$, $(E, D_i)|(U_i \cap U_j)$ and $(E, D_j)|(U_i \cap U_j)$ are isomorphic as flat bundle families.
- For each $p \in P$, $(E|(\{p\} \times X), D_p)$ is a flat bundle on X.
- The $(E|(U_i \times X), D_i)$ extend to continuous families of flat bundles parametrized by $(P \cap \overline{U_i}) \cup U_i$, that are slicewise isomorphic on $P \cap \overline{U_i} \cap \overline{U_j}$.

will be called a sporadic family of harmonic bundles on X parametrized by U. Now, consider the tuple $(U, P, \{U_i\}_{i \in I}, E, \{D_i\}_{i \in I})$, and remove the harmonicity requirement over the U_i. We will call such objects sporadic families of flat bundles.

In particular, such families yield continuous maps into the Dolbeault and de Rham coarse moduli spaces. This motivates us to consider a further generalization of the de Rham moduli stack that incorporates sporadic families, and I hope to address the following questions:

Question 2.1. Do sporadic families of flat bundles organize into a stack $\mathcal{M}_{\mathrm{dR}}^{\mathrm{sporadic}}(X)$? Is this stack geometric? Gluings on the distinguished closed subsets of parameter spaces are the crux.

Question 2.2. Sporadic harmonic bundle families should assemble into a prestack $\mathcal{M}_{\mathcal{H}}^{\mathrm{sporadic}}(X)$ with forgetful maps to $\mathcal{M}_{\mathrm{Dol}}(X)$ and $\mathcal{M}_{\mathrm{dR}}^{\mathrm{sporadic}}(X)$. Can the stackified extension completion technique of [1] be adapted to this setting? How much of the polystable locus in $\mathcal{M}_{\mathrm{Dol}}(X)$ does the essential image of the stackified extension completion of $\mathcal{M}_{\mathcal{H}}^{\mathrm{sporadic}}(X)$ cover?

Question 2.3. How does regularity of the isomorphisms of flat bundle families in the parameter affect the coverage of the polystable locus?

3 Categorical and Derived Enhancements. Sporadic families of flat bundles demonstrate the need for singular gluing of objects, and derived geometry likely has a role here that I would like to explore. Repeating the methods of [1] with C^∞–schemes in place of manifolds, we should obtain geometric derived stacks $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf}}(X)$, $\mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf}}(X)$ over C^∞–schemes parametrizing perfect complexes of Higgs bundle and flat bundle families respectively. We should also have a derived prestack $\mathfrak{M}_{\mathrm{wk}\mathcal{H}}^{\mathrm{perf}}(X)$ of perfect complexes of weak harmonic families. There should be forgetful maps $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf}}(X) \longleftarrow \mathfrak{M}_{\mathrm{wk}\mathcal{H}}^{\mathrm{perf}}(X) \longrightarrow \mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf}}(X)$. Adapting the extension completion construction should give an equivalence of the essential images $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf,wk}\mathcal{H}} \subset \mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf}}$ and $\mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf,wk}\mathcal{H}} \subset \mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf}}$.

Question 3.1. How much of the semistable and polystable loci in $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf}}(X)$ are covered by $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf,wk}\mathcal{H}}$? What characterization can we give the locus in $\mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf}}(X)$ covered by $\mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf,wk}\mathcal{H}}$?

A further categorical generalization is possible. In [3], we constructed simplicial algebraic stacks $\mathcal{M}_{\mathrm{Dol}}^{\bullet}(X), \mathcal{M}_{\mathrm{dR}}^{\bullet}(X)$ whose n–th degree parametrizes Δ^n–shaped diagrams of families of Higgs bundles and flat bundles respectively on a smooth complex projective variety X. There are also derived algebraic geometric counterparts of these [7, §4] and we can expect analogous simplicial derived C^∞–stacks $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf},\bullet}(X), \mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf},\bullet}(X)$. They should also form ∞–category objects internal to the ∞–category of derived C^∞–stacks, perhaps similar to the 2–Segality in [5, Lemma II.4.1]. We should have a simplicial derived prestack $\mathfrak{M}_{\mathrm{wk}\mathcal{H}}^{\mathrm{perf},\bullet}(X)$ for weak harmonic families, with simplicial mappings $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf},\bullet}(X) \longleftarrow \mathfrak{M}_{\mathrm{wk}\mathcal{H}}^{\mathrm{perf},\bullet}(X) \longrightarrow \mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf},\bullet}(X)$, by arguments similar to the non-derived case. In this context, the questions I hope to investigate are:

Question 3.2. Can extension completion give an equivalence, in some sense appropriate for simplicial derived C^∞–stacks, between simplicial substacks $\mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf,wk}\mathcal{H},\bullet}(X) \subset \mathfrak{M}_{\mathrm{Dol}}^{\mathrm{perf},\bullet}(X)$ and $\mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf,wk}\mathcal{H},\bullet}(X) \subset \mathfrak{M}_{\mathrm{dR}}^{\mathrm{perf},\bullet}(X)$. How much of the polystable and semistable loci do these cover?

Question 3.3. How do sporadic harmonic bundle and flat bundle families fit into this picture?

4 Alternatives to the Correspondence. The harmonicity condition is what forces our stack-level NAH correspondence to fail on parts of the polystable locus. Thus, I wish to consider the possibility of correcting the mechanism. For instance:

Question 4.1. Can we modify the formula translating between a Higgs field and a flat connection using a metric, and adapt the analysis of [10] to produce these metrics with regularity in the parameter for all polystable families?

Since maintaining the geometricity of the NAH correspondence requires increasingly complicated machinery on larger loci, it is also interesting to consider moving away from mediation by harmonic metrics altogether. The non-Abelian Hodge filtration stack provides motivation

for this. This is the stack whose objects are of the form (U, λ, E, D), where U is a smooth manifold, $\lambda : U \longrightarrow \mathbb{C}$ is a continuous function, $E \longrightarrow U \times X$ is a complex bundle and D is a λ–$d_{X/U}$–connection in the sense of [1, §2.4]. Call such an object a Hodge family. Given a family of harmonic bundles $(E, \overline{\partial}_{E/U}, \theta, \partial_{E/U,h}, \theta^{\dagger h})$ for a harmonic metric h continuous in the parameter, we have a standard Hodge family defined by $\nabla_\lambda = \overline{\partial}_{E/U} + \partial_{E/U,h} + \lambda^{-1}\theta + \lambda\theta^{\dagger h}$. It is natural to ask: given a family of Higgs bundles (E_0, D_0''), is there another method to produce a Hodge family $\mathcal{F} := (U, \lambda, E, D)$ such that $\mathcal{F}_0 := (E, D)|\lambda^{-1}(0)$ is equivalent to (E_0, D_0'')? Bertrand Eynard and Raphaël Belliard have discussed with me a method to produce such a Hodge family, for possibly non-harmonic bundles, using topological recursion. Given this exciting possibility, I hope to investigate the following questions:

Question 4.2. How functorial is this construction of Hodge families? Can we organize it into a mapping of stacks $\mathcal{M}_{\mathrm{Dol}}(X) \longrightarrow \mathcal{M}_{\mathrm{Hodge}}(X)$ as the one we constructed in [2, §13]?

Question 4.3. Does the mapping $\mathcal{F} \longmapsto \mathcal{F}_1$ assemble to a mapping of stacks $\mathcal{M}_{\mathrm{Hodge}}(X) \longrightarrow \mathcal{M}_{\mathrm{dR}}(X)$?

Question 4.4. What is the derived enhancement of this picture? What is the enhancement in terms of simplicial derived stacks?

Question 4.5. If we can construct this chain of maps $\mathcal{M}_{\mathrm{Dol}}(X) \longrightarrow \mathcal{M}_{\mathrm{Hodge}}(X) \longrightarrow \mathcal{M}_{\mathrm{dR}}(X)$, is it an equivalence when restricted to a substantial locus of $\mathcal{M}_{\mathrm{Dol}}(X)$? Ultimately, does this increase the locus coverage compared to the NAH correspondence?

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